

Guided Inquiry: Fundamentals of Liquid-Liquid Extraction (LLE)

Using the Hunter-Nash Ternary Diagram Method

Introduction

In this activity, you will use an interactive ternary phase diagram to explore the principles of liquid-liquid extraction. Extraction is a mass-transfer process where a solute is transferred from one liquid phase (the carrier) to another immiscible liquid phase (the solvent).

Instructions: Open the interactive LLE app. Under "System Parameters," select **Water - Acetone - MIBK**. In this system:

- **Carrier (A):** Water
- **Solute (B):** Acetone (the chemical we want to extract)
- **Solvent (C):** MIBK (Methyl Isobutyl Ketone - the chemical we are using to "wash" the acetone out of the water)

Part 1: Anatomy of the Ternary Diagram

Before separating chemicals, we must understand how they mix.

1. **The Pure Components:** Locate the three corners of the triangle. Where does a mixture containing 100% pure Water plot?
2. **The Binodal Curve:** Look at the heavy black curve forming a "dome." This is the *binodal curve*.
 - If a mixture's overall composition plots **outside/above** the dome, what physical state is it in? (Hint: Does it separate, or is it fully miscible?)
 - If a mixture's overall composition plots **inside/below** the dome, it is in the "two-phase region." What happens physically to a mixture poured into a beaker if its overall composition lies inside this dome?

Part 2: Equilibrium and Tie Lines

Inside the dome, a mixture will split into two distinct liquid layers.

1. Notice the gray background lines connecting the left side of the dome to the right side. These are **Tie Lines**.
2. The left side of the curve (rich in Water) is the **Raffinate** phase. The right side (rich in MIBK) is the **Extract** phase.
3. If you have an overall mixture that lands exactly on the middle of a tie line, what are the exact compositions of the two liquid layers that form? How does the tie line help you find this?
4. *Observation:* Are the tie lines perfectly horizontal? Because they slope upwards toward the right, does Acetone prefer to dissolve in Water or MIBK at equilibrium?

Part 3: The Mixing Point (M) and the Lever Rule

Let's introduce our feed and solvent.

- Set **Feed Solute** ($x_{F,B}$) to **30%** (0.30).
 - Set **Solvent/Feed Ratio** to **1.5**.
1. Locate point **F** (green, Feed) and point **S** (purple, Solvent). Locate point **M** (orange). What does point **M** physically represent in an extraction process?
 2. Notice that **M** lies exactly on a straight line connecting **F** and **S**.
 - *Experiment:* Slide the **Solvent/Feed Ratio** up to 3.0. Which point does **M** move closer to?
 - Slide it down to 0.5. Which point does **M** move closer to?
 - *Conclusion:* Formulate a rule (The Lever Rule) relating the position of **M** on the line segment $\overline{FS}$ to the amounts of Feed and Solvent mixed together.
 3. *Experiment:* Keep sliding the **Solvent/Feed Ratio** higher. Eventually, the app will throw an error: "Mixing point (M) is in the single-phase region." Look at the diagram. Why is extraction physically impossible if **M** is pushed outside the dome?

Part 4: Operating Lines and Stepping off Stages

Reset the parameters: **Feed = 30%**, **S/F Ratio = 1.5**, **Target Raffinate = 5%**.

Zoom in on the diagram to clearly see the alternating red and blue lines.

1. **The Difference Point (Δ):** Locate the dark blue point Δ floating outside the triangle. This point represents a fixed material balance constraint for the entire cascade of extraction stages.
2. **The Staging Process:** You will see a "staircase" pattern between the two sides of the dome.
 - The **Red Lines** are Tie Lines. They represent liquid-liquid *equilibrium*. What physical piece of equipment allows two liquids to reach equilibrium?
 - The **Blue Lines** point toward Δ . These are *Operating Lines*, representing the law of conservation of mass as fluids flow between stages.
3. Count the number of red lines. Compare this to the "Ideal Stages Needed" reported in the Analysis Results panel. What does one "step" (one red line + one blue line) represent in the real world?

Part 5: Pinch Points and Minimum Solvent

The amount of solvent you use heavily dictates the cost of operating the chemical plant. Let's see what happens if we are stingy with our solvent.

1. Leave the Feed at 30% and Target Raffinate at 5%.
2. Slowly decrease the **Solvent/Feed Ratio** slider from 1.5 downwards. Watch point Δ and

- the blue operating lines. What happens to the position of Δ as you use less solvent?
3. Keep decreasing the ratio until you reach about **0.6**. An orange warning will appear, and eventually a red error: *"Pinch point reached."*
 4. Look closely at the tie lines (red) and operating lines (blue) near the Feed point (F).
 - What geometric property occurs between an operating line and a tie line that creates a "pinch"? (Hint: Look at their slopes/parallelism).
 - If you reach a pinch point, how many stages are required to reach your target purity?
 5. **Engineering Application:** Based on this observation, what is the trade-off an engineer must balance when choosing a Solvent-to-Feed ratio for a new chemical plant? (Think about the cost of purchasing/recycling solvent vs. the physical size and cost of the extraction column).